

Muhammad Ali Alam

M.Sc. Computer Science · Bioinformatics & Machine Learning

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linkedin.com/in/muhammad-ali-alam | pypi.org/project/genobench

PROFILE

Seeking a Master's thesis position in Computational Biology to bridge ML expertise with genomic sequence modelling. Machine Learning researcher at the University of Siegen specializing in sequential deep learning and multimodal AI, with a focused transition into Computational Genomics. Author of **GenoBench** — a published Python framework (PyPI, July 2026) for joint biological and hardware benchmarking of Genomic Foundation Models (HyenaDNA, DNABERT-2). Experienced in HPC-scale model development (SLURM), transformer architectures, and XAI. Backed by two recent papers (CoDIT, IEEE) in multimodal AI.

RESEARCH PROJECTS

ECHO — Efficient Clinical Health Observer — University of Siegen - [repository](#) May 2026 – Present

- **Benchmarking & Modeling:** Conducted a multi-paradigm evaluation of audio-only depression detection on DAIC-WOZ, comparing advanced SSL embeddings (WavLM, HUBERT) with classical feature sets via linear probing and fine-tuning.
- **Reduced GPU VRAM consumption by 100x** (eliminating OOM failures to a stable 9.35 GB) by engineering a 1D depthwise-separable convolutional downsampling layer, effectively resolving $O(N^2)$ Transformer memory bottlenecks.
- **Accelerated speech preprocessing by 64x** (reducing per-session latency from 15 minutes to 14 seconds) by migrating diarization and embedding extraction pipelines to CUDA GPUs, executing dataset-wide preprocessing in under 7 minutes via SLURM parallel job arrays.

A Multimodal Framework for Depression Detection — University of Siegen - [repository](#) Apr 2025 – Mar 2026
WavLM + MentalRoBERTa + Cross-Modal Transformer with XAI on E-DAIC

- Developed a biometric identity gating protocol (ResNet34 speaker embeddings) achieving **100% pure patient speech** by eliminating interviewer leakage from clinical recordings.
- Architected a session-level **Cross-Modal Transformer** with bi-directional co-attention to dynamically weight acoustic prosody against semantic clinical themes across 7–33 minute sessions.
- Applied **Multi-Task Learning (MTL)** for PHQ-8 symptom regression with label smoothing to prevent overfitting on a constrained clinical dataset.
- Built an end-to-end **XAI pipeline** (Attention Rollout + Gemini 3.1 Pro) synthesising raw neural weights into human-readable clinical narratives via a Streamlit diagnostic dashboard.

OPEN SOURCE SOFTWARE

GenoBench v1.0.0 — PyPI · MIT License · github.com/M-Ali-Alam/GenoBench Released Jul 1, 2026
A joint biological and hardware benchmarking framework for Genomic Foundation Models (GFMs).

- **Joint profiling:** Combines biological metrics (MCC, AUPRC) with GPU hardware metrics (peak VRAM, CUDA-event inference latency) under a standardised linear probing protocol on frozen sequence embeddings.
- **Built-in models & tasks:** HyenaDNA-tiny and DNABERT-2; Genomic Benchmarks datasets (human vs. worm classification, human enhancer identification). Reproducible seeding, structured JSONL output schema.
- **Planned extensions:** Support for nucleotide transformer models, chromatin accessibility prediction tasks, and multi-GPU benchmarking.

PUBLICATIONS

- [1] Ugonna Oleh, Umair Talib, **Muhammad Ali Alam**, Roman Obermaisser. "Leakage-Resilient Diarization and Feature Extraction Framework for Robust Multimodal Mental Health Corpora." *2026 12th International Conference on Control, Decision and Information Technologies (CoDIT)*.
- [2] Umair Talib, Ugonna Oleh, **Muhammad Ali Alam**, Roman Obermaisser. "The Glass Box Psychiatrist: Grounding LLM Clinical Narratives in Multimodal Transformer Attention Rollouts." *IEEE COINS 2026*.

SKILLS

Languages	Python · R · TypeScript · SQL (PostgreSQL) · Bash/Linux
ML / AI	PyTorch · TensorFlow · HuggingFace · pandas · NumPy

DevOps

SLURM Docker · AWS (Lambda, EC2, S3) · GitHub Actions · CI/CD

Bioinformatics

Biopython, PyRanges, FASTA, FASTQ, k-mer tokenization, positional encoding for DNA

PROFESSIONAL & TEACHING EXPERIENCE

AI Mentor — AuratTech · DataCamp Programme

Jul 2026 – Sep 2026

- Mentoring **45 students** through DataCamp's "AI Engineer for Data Scientists Associate" certification; twice-weekly 2-hour live Zoom sessions across 8 weeks.
- Designed structured session curriculum, Git-first onboarding, and triage-based progress tracking; completed Training of Trainers onboarding.

Full Stack Software Developer — CreditBook — Fintech Platform

Jul 2022 – Jun 2023

- Built and maintained production APIs and frontend features serving **over 1 million users**; integrated OTPless SDK reducing infrastructure costs by over **1M PKR**.
- Led A/B testing via Firebase, improving feature adoption by **18.5%**; automated push notifications via MoEngage SDK.

Teaching Assistant — IBA, Karachi

Aug 2021 – May 2022

- Led tutorials and Q&A for **70+ students** in data structures, algorithms, and operating systems; assisted in assignment design emphasising code quality.

EDUCATION

M.Sc. Computer Science — University of Siegen

Oct 2023 – Present

ERASMUS+ exchanges: Solution Design at Univerza v Mariboru, Slovenia (Aug 2025) · Makers 5.0 at VILNIUS TECH, Lithuania (Apr 2024)

B.Sc. Computer Science — IBA, Karachi

Aug 2018 – May 2022

LANGUAGES

English — Fluent (C1)

German — Intermediate (B1)

Urdu / Hindi — Native